BlackArch Linux on a Dell laptop, driven over SSH - step by step

Source: blackarch.org/downloads.html, blackarch.org/blackarch-install.html and the official PDF guide (blackarch-guide-en.pdf from blackarch.org/guide.html, section 2.1.5). Dell-specific notes are marked **Dell**.

0. Which ISO and why (read this once)

ISO	Size	Date	Verdict
Full	22 GB	2023-04-01	BlackArch's own page: "highly discouraged ... many errors and conflicts". Frozen 2023 packages.
Slim	5.5 GB	2023-05-01	GUI installer, a curated subset of tools.
Netinstall	815 MB	2023-04-01	**Official recommendation.** Bootstrapper only; everything is pulled fresh from the repos.

"Most complete" = netinstall + `pacman -S blackarch` after first boot. That installs **every** tool currently in the repo (2800+, newer than anything on the full ISO) with no 2023-vs-2026 conflicts.

1. Write the stick (Windows PC)

```
cd C:\Projects\BlackArch
.\Write-IsoToUsb.ps1 -IsoPath "$env:USERPROFILE\Downloads\blackarch-linux-netinst-2023.04.01-x86_64.iso" -DiskNumber 2 -ExpectedSha1 df2dfcfafa1a8f9bab8afd9aa6b788de00e03e09
```

It elevates itself, checks the SHA1, shows the target disk, asks you to type `2`, wipes the stick and dd's the ISO. When Windows says "format this disk?" click **Cancel**. Eject.

> Check the disk number first with `Get-Disk` - the 128 GB stick was disk 2 in our case.

2. Dell BIOS (F2 at the Dell logo)

- **Secure Boot: OFF** (the BlackArch ISO is not signed).
- SATA/NVMe operation: **AHCI/NVMe**, not RAID/RST (Linux won't see the SSD under RST).
- Boot the stick: power on, tap **F12**, pick the **UEFI** entry for the USB stick.
- At the ISO menu choose **BlackArch Linux (x86_64)**.

3. Network - do this BEFORE starting the installer

Dell: the laptop has no Ethernet port and the installer script does not do Wi-Fi. Easiest: plug a USB-C Ethernet adapter / dock. Otherwise connect Wi-Fi by hand, then choose **Skip** at the installer's network step:

```
ip a                              # find the wifi interface name (wlan0 / wlp0s20f3)
iwctl                             # if present
  station wlan0 scan
  station wlan0 get-networks
  station wlan0 connect "YourSSID"   # asks for the password
  exit
# if iwctl is missing, try:  wifi-menu
ping -c 2 archlinux.org
```

Log in as `root`, password `blackarch` (live ISO defaults).

4. Fix the 2023 keyrings (prevents "signature is unknown trust" failures)

The ISO is from 2023; refresh the keys before installing anything:

```
pacman -Sy archlinux-keyring blackarch-keyring
pacman-key --init
pacman-key --populate archlinux blackarch
```

5. Run the installer

Run it over SSH from the Windows PC (tmux/screen from today's repos will not run on the 2023 live system, so keep the dock cable plugged in; if the SSH tab dies, reconnect and start the installer again).

```
pacman -Sy blackarch-installer     # get the current installer (1.2.25 on 2026-10-06)
ssh -t root@<live-iso-ip> blackarch-install
```

Answers for installer 1.2.25 (prompts appear in this order):

Prompt	Answer
Choose an installation mode	**1** - Install from repository using pacman
Output mode (verbose/quiet)	**1** verbose, so errors are visible
Set locale	Enter (`en_US.UTF-8`)
Keymap	Enter (`us`)
Set your hostname	e.g. `blackarch-laptop`
Network interface	`eth0` (the dock)
Network mode	**1** Auto DHCP
Install with Windows/Other OS (dualboot)	**n**
Full encrypted root (LUKS)	your call. `y` = passphrase at every boot, data safe if laptop is lost.
Please choose a device	`nvme0n1` (the 1 TB NVMe). **This erases the SSD.** Never `sda`, that is the USB stick.
Create partitions with cfdisk	`y`
Start with an in-memory zeroed partition table	`y` (wipes the old Omarchy layout)
cfdisk: label type	**gpt**
cfdisk: partition 1	New -> `1G` -> Type -> **EFI System**
cfdisk: partition 2	New -> `32G` -> Type -> **Linux swap** (matches RAM, allows hibernate)
cfdisk: partition 3	New -> Enter (rest of disk) -> leave as Linux filesystem; **Write** -> `yes` -> **Quit**
EFI System partition	`1` (the installer expands it to `/dev/nvme0n1p1`; fat32 is automatic)
Root partition / filesystem	`3` / Enter (ext4)
Swap partition	`2`
Partition table correct	`y`
Timezone (Zone/SubZone)	`America/New_York` (adjust)
root password, user name + password	set both; don't run as root day to day

Prompt	Answer
Select a mirror number	Enter for default
Setup X11 + window managers	**y** (installs LXDM login screen + window managers)
Choose window managers [6]	Enter = all five (Awesome, Fluxbox, i3, Openbox, Spectrwm). Pick Openbox at login for a normal mouse-driven desktop.
VirtualBox / VMware guest tools	**n** (real hardware)
Setup BlackArch Linux tools	**n** - do it after first boot (step 7) over SSH where errors can be handled

Reboot, remove the stick.

5b. Unattended run (what we actually did on 2026-10-06)

`Tools\drive-install.sh` runs the installer inside a pty on the live ISO and types every answer (partitions were made first with `sfdisk`: 1G EFI / 32G swap / rest root, LUKS on root). Launch from Windows:

```
scp C:\Projects\BlackArch\Tools\drive-install.sh root@<live-iso-ip>:/root/
ssh root@<live-iso-ip> "sed -i 's/\r$//' /root/drive-install.sh; chmod +x /root/drive-install.sh; setsid nohup /root/drive-install.sh blackarch-laptop pentester America/New_York nvme0n1 > /root/driver.out 2>&1 < /dev/null &"
```

Progress: `ssh root@<live-iso-ip> cat /root/driver.log`. Full installer output: `/root/install.log`. It sets **temporary** secrets, all `blackarch`: LUKS passphrase, root password, user password.

6. First boot - change the three temporary secrets FIRST

At the boot prompt "Enter passphrase" type `blackarch`. Log in as `pentester` / `blackarch`, then:

```
passwd                                    # your user password
sudo passwd root                          # root password
sudo cryptsetup luksChangeKey /dev/nvme0n1p3  # disk passphrase: asks old (blackarch), then new twice
```

Verify the new disk passphrase before you forget the old one: `sudo cryptsetup open --test-passphrase /dev/nvme0n1p3`.

The installed system came up at (the live ISO had .172). The Windows PC's SSH key was pre-installed for root and pentester, so from Windows: `ssh pentester@<laptop-ip>`.

Then:

```
sudo pacman -Syyu            # full system update
```

Dell: if Wi-Fi is missing after install: `sudo pacman -S iwd` then `sudo systemctl enable --now iwd` (or `networkmanager` + `nmtui`). Firmware for the Intel AX211 card is in `linux-firmware`, already pulled in.

7. Install EVERYTHING

```
sudo pacman -S blackarch     # all 2800+ tools; press Enter to accept defaults. Needs
~40-60 GB.
```

Takes a while. If it stops with "failed to commit transaction / conflicting files", the official fix is:

```
sudo pacman -Syyu --needed --overwrite='*' blackarch
```

Or install by category instead: `pacman -Sg | grep blackarch` then `sudo pacman -S blackarch-<category>`.

Done 2026-10-06 16:56-18:29 MDT: 2,855 tools / 6,028 packages, 25 GB download, 130 GB on disk. 29 tools had non-fatal post-install hook failures (old Ruby gems / pinned Python deps); the packages are installed, those tools may need their own setup the first time. The list is in the pacman log (`grep "command failed" /var/log/pacman.log`).

7b. Daily-driver desktop, drivers, firmware, performance (what we ran)

`Tools\post-tools-desktop.sh` (run as root after the tools): GNOME + GDM login screen, Ubuntu-style dock (`gnome-shell-extension-dash-to-dock` from the AUR via paru), tray icons, Firefox, NetworkManager, PipeWire audio, Bluetooth, `nvidia-open` + `nvidia-utils` + `nvidia-prime` (RTX laptop GPU), Intel media driver, fwupd, power-profiles-daemon set to **performance**, weekly `fstrim.timer`.

Lessons: - The Arch package is **`nvidia-open`** now (not `nvidia`) for RTX cards. A wrong name aborts the whole pacman batch. - Always `sudo pacman -Syu` **before** installing anything. The repo kernel moved 7.2.8 -> 7.2.9 during our session, so the NVIDIA module was built for a kernel that was not running yet. Full

upgrade + reboot fixed it. - Run the GPU on demand: `prime-run <program>` sends that program to the NVIDIA card; the Intel GPU drives the desktop. `hashcat -I` lists both once `opencl-nvidia` is installed. - After switching to NetworkManager the laptop got a new DHCP address (). Use `ssh pentester@blackarch-laptop` or check `ip -br a` on the laptop.

7c. Choosing a desktop (several are installed, pick at the login screen)

Installed on blackarch-laptop: **GNOME** (Wayland only), **KDE Plasma** (Wayland + X11), **Cinnamon** (X11 and Wayland), **Xfce**, plus the five bare window managers from the installer (awesome, fluxbox, i3, openbox, spectrwm - keyboard-driven, skip them). To switch: log out, click your name at the GDM login screen, then the **gear icon bottom-right**, pick a session, enter the password. GDM remembers the last choice per user.

Desktop	Feel	BlackArch tool menu (categories)
KDE Plasma	Windows-like taskbar + start menu, most configurable, good on NVIDIA	yes (start menu)
Cinnamon	Linux Mint; simplest, bottom panel + menu	yes
GNOME	Ubuntu-like dock + Activities overview	no (flat app grid; add the ArcMenu extension for categories)
Xfce	lightweight, traditional	yes

"X11" vs "Wayland" entries: same desktop, different display system. Wayland is newer and smoother; X11 is the one almost every pentest GUI tool was written for (screen capture, Java apps, Burp, Wireshark). On this laptop prefer the **X11** entry when a tool misbehaves. The Xfce Wayland entry needs `labwc`, which is not installed, so it just bounces back to the login screen.

Install/remove desktops with pacman, e.g. `sudo pacman -Rns xfce4 xfce4-goodies` once you have settled.

8. If the installer dies at the ping check

The script aborts when `ping github.com` fails (firewalls that drop ICMP). Official workaround: edit `/usr/bin/blackarch-install`, find the block starting `if ! ping -c 1 github.com` (around line 545) and delete through its `fi`, then rerun `blackarch-install`.

Plan B - plain Arch first, BlackArch on top

If `blackarch-install` won't cooperate on this laptop, install vanilla Arch (archinstall or Omarchy) and then, as root:

```
curl -O https://blackarch.org/strap.sh
echo d338a4bb95d9e09f97508da68ac9e17d963b85f6 strap.sh | sha1sum -c
chmod +x strap.sh && sudo ./strap.sh
sudo pacman -Syu && sudo pacman -S blackarch
```

Result is the same repo and the same tools - that is literally what the netinstall path does.

9. Dell laptop hardware notes (Linux has no "driver pack" step)

Linux drivers live inside the kernel and the `linux-firmware` package. Both are installed by the installer automatically, so there is nothing to slipstream onto the stick (the stick is a read-only dd image anyway). The only things to add are done AFTER install, with pacman:

Hardware	Status	After install
Thunderbolt dock Ethernet (Realtek)	in-kernel `r8152` / `r8169` - works in the live ISO, pick **Auto DHCP**	nothing
Intel AX211 Wi-Fi / Bluetooth	in-kernel `iwlwifi` + firmware	`sudo pacman -S networkmanager && sudo systemctl enable --now NetworkManager` then `nmtui`
Intel Iris Xe iGPU	in-kernel `i915`	`sudo pacman -S mesa intel-media-driver`
NVIDIA RTX laptop GPU (this laptop)	boots on `nouveau`; proper driver is `nvidia-open`	`sudo pacman -Syu nvidia-open nvidia-utils nvidia-prime opencl-nvidia` then reboot; verify with `nvidia-smi`
Touchpad, audio, webcam, USB-C	in-kernel	nothing (audio: `sudo pacman -S pipewire pipewire-pulse wireplumber`)
Fingerprint reader (Goodix)	not supported on Linux	skip

Dock tips: in BIOS keep **Thunderbolt > Enable Thunderbolt Boot Support** ON and security level "No security" or "User Authorization"; plug the dock in before powering on so the live ISO sees it. Check with `ip a` - the dock NIC shows as `enp...` or `eth0`; `ping -c 2 archlinux.org` must answer.

10. Talking to Claude while installing

Keep this Claude Code session open on the Windows PC. Photograph the Dell screen with your phone and paste the photo into the chat; that is enough to read error messages. Or type the exact prompt text.